NOT MEASUREMENT SENSITIVE

MIL-C-81309E
AMENDMENT 3
26 November 1997
SUPERSEDING
AMENDMENT 2
31 July 1996

MILITARY SPECIFICATION

CORROSION PREVENTIVE COMPOUNDS, WATER DISPLACING, ULTRA-THIN FILM

This amendment forms a part of MIL-C-81309E, dated 31 March 1993, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.2.2, line 3: Delete "Grade 22 - Hydrochlorofluorocarbons (HCFC) propellant." and substitute "Grade CO_2 - Carbon Dioxide (CO_2) propellant."

PAGE 2

* 2.1.1: Add the following under "FEDERAL SPECIFICATIONS":

TT-N-95 - Naptha, Aliphatic

PAGE 4

* 2.2: Add the following under "AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)":

ASTM-D93 - Flash Point by Pensky-Martens Closed Cup Tester IP Designation: D34/88, Standard Test Methods for (DoD Adopted).

PAGE 5

3.2.1.1: Delete in entirety and substitute:

"3.2.1.1 Grade CO_2 propellant. The propellant for grade CO_2 shall consist of carbon dioxide. The propellant shall contain no chlorinated solvents, hydrochlorofluorocarbons, or fully halogenated chlorofluorocarbons."

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* 3.3: Delete lines 3 and 4 and substitute: "Commander, Naval Air Warfare Center Aircraft Division, Attn.: Code 4.3.4.1 Building 2188, 22347 Cedar Point Road, Unit 6, Patuxent River, MD 20670-1161."

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* 3.5.2: Delete in entirety and substitute:

"3.5.2 Fill. When tested as specified in 4.7.12.2, the pressurized cans containing the compound shall have a minimum net weight as shown below:

Type, class, and grade	Minimum net weight
Type II, class 2, grade CO_2	11.0 ounces
Type II, class 2, grade 134A	14.5 ounces
Type III, class 2, grade CO_2	11.0 ounces
Type III, class 2, grade 134A	14.0 ounces

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* Table I, first column:

Delete "Synthetic sea water-sulfurous acid spray" and substitute "Acid salt fog protection".
Delete "Synthetic sea water displacement" and substitute "Water displacement".

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* 4.3.1, line 6, after "the": Delete the remainder of sentence and substitute: "Commander, Naval Air Warfare Center Aircraft Division, Attn.: Code 4.3.4.1 Building 2188, 22347 Cedar Point Road, Unit 6, Patuxent River, MD 20670-1161."

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* Table II, first column:

Delete "Synthetic sea water - sulfurous acid spray" and substitute "Acid salt fog protection".
Delete "Synthetic sea water - displacement" and substitute "Water displacement".
Delete "Flame projection" and substitute "Fill".
Delete "Fill" and substitute "Performance".
Delete "Performance" and substitute "Flame projection".

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* Table III, first column:

Delete "Synthetic sea water - sulfurous acid spray" and substitute "Acid salt fog protection". Delete "Synthetic sea water replacement" and substitute "Water displacement".

* 4.6.1, line 4: Delete "synthetic sea water-sulfurous acid spray" and substitute "acid salt fog protection".

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* 4.6.2, line 2: Delete "synthetic sea water-sulfurous acid spray" and substitute "acid salt fog protection".

* 4.6.4, line 3: Delete "(see Figure 1)".

* 4.6.5: Delete the last sentence and substitute: "After removal, they shall remain in a vertical position for 24 hours, under the atmospheric conditions of 4.5, and in a draft, dust, and fume-free atmosphere."

* 4.7.1 and 4.7.1.1: Delete in entirety and substitute:

"4.7.1 Minimum flash point. The flash point shall be determined in accordance with ASTM-D93."

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4.7.2: Delete the entire test procedure and substitute:

"4.7.2 Acid salt fog protection.

4.7.2.1 Conditioning of compound sample. A wide-mouth, quart jar containing 800 mL of the compound shall be sealed and exposed to the following cycle four times without physical disturbance of the compound: 8 hours at 54.4 ±1.1°C (130 ±2°F) followed by 16 hours at –40 ±1.1°C (–40 ±2°F). At the end of this cyclic exposure, the contents of the sealed jar shall be allowed to remain at 23.9 ±2.8°C (75 ±5°F) for 24 hours.

4.7.2.2 Test solution. The spray solution shall consist of a solution made by adding 2 mL of sulfurous acid (6.4 percent assay as SO_2) to each liter of synthetic sea water. The pH shall be measured and shall be 3.4 ±0.1. If it is not, additional acid or synthetic sea water shall be added to adjust the pH to this range. The synthetic sea water shall be prepared by adding

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50 grams of sodium chloride (NaCl), 22 grams of magnesium chloride ($MgCl_2 \cdot 6H_2O$), 3.2 grams of calcium chloride ($CaCl_2 \cdot 2H_2O$), and 8 grams of sodium sulfate (Na_2SO_4) to a liter of distilled or demineralized water. One liter of fresh solution shall be used for each cycle.

4.7.2.3 Test procedure. The apparatus used in the test shall conform to figures 1, 2, and 3. Test disks, prepared as specified in 4.6, shall be coated with the conditioned compound of 4.7.2.1. The disks shall be hung vertically and then dried for 24 hours under standard laboratory conditions (see 4.5). After inserting the specimens, the turntable shall be started and the air flow to the nozzle regulated to effect a fine spray. The pickup tube supplying the nozzle shall have an inside diameter (ID) of 1 mm. At the end of approximately 2 minutes of spraying, 35 mL of water, which has been cooled to 1.7°C (35°F), shall be added to each aluminum water cup to cause a fine condensation to form on the surface of the disks. Spraying shall continue for 1 hour; the rate of flow shall be approximately 1500 mL/hour. At the end of the hour, the spray shall be stopped, the aluminum cups removed, the jar removed from under the turntable, and the disks allowed to dry in place at room temperature for 3 hours. (During this time, to avoid condensate runoff washing over the test specimens, excessive condensation shall not be allowed to accumulate on the OUTSIDE of the water cups.) Another cycle shall then be started with fresh test solution. Two cycles shall be conducted in one day. After the required number of cycles, the disks shall be cleaned in mineral spirits conforming to type II of P-D-680 or type III of TT-T-291, after which they shall be examined for any visible corrosion (see table I)."

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Add the following paragraph:

"4.7.3 Water displacement. An uncoated panel prepared as specified in 4.6 shall be placed so that one 2-inch end shall be raised 1 inch above a horizontal surface (a 25 percent incline). The panel shall then be sprayed with the unacidified synthetic sea water of 4.7.2.2 so that its entire upper surface is covered with tiny droplets. Within 1 minute after spraying, 1 mL of the test compound shall be poured along the upper 2-inch edge of the panel and allowed to run slowly down so as to completely cover the test panel. After another minute, a second 1-mL sample of the test compound shall be poured and allowed to run down in a like manner. After waiting an additional minute, the panel shall be raised to a vertical position for 1 minute and then be placed flat, test side up, above distilled water at 22°C (72°F) in a closed desiccator. After 4 hours, the panel shall be removed, cleaned with mineral spirits, and evaluated for the presence of corrosion (see table I)."

* 4.7.5.1, line 2 and line 4, after "of": Delete "xylene" and substitute "aliphatic naphtha conforming to TT-N-95".

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4.7.10.1, line 4, after "at": Delete the remainder of paragraph and substitute:

"77 ±1°C (170 ±2°F) for 15 minutes. There shall be no sedimentation or evidence of chemical precipitation or reaction immediately after the 15-minute period at the elevated temperature or after 24 hours at 25 ±3°C (77 ±5°F) (see table I). A slight separation of liquid shall be acceptable."

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* 4.7.12.2: Delete the last sentence and substitute: "The net difference shall be as specified in 3.5.2."

PAGE 21

* 6.1.2, line 1: Delete "Type II" and substitute "Type III".

* 6.3: Delete the last sentence and substitute: "Information pertaining to qualification of products may be obtained from the Commanding Officer, Naval Air Warfare Center Aircraft Division Patuxent River, Code 4.3.4.1, Building 2188, 22347 Cedar Point Road, Unit 6, Patuxent River, MD 20670-1161."

PAGE 22

* 6.5, line 14, after "Grades": Delete "22" and substitute "CO_2".

* 6.6: Delete part number breakout in entirety and substitute:

M81309 -	X	X	X
Specification identification	Type (see 1.2.1)	Class (see 1.2.1)	Grade (see 1.2.2)
	A = Type II	1 = Class 1	CO2 = Grade CO_2
	B = Type III	2 = Class 2	134A = Grade 134A

Example: A type II, class 2, grade CO2 compound is designated as M81309-A2CO2

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* Table IV, footnote 1/: Delete "1.1.1" and substitute "1.2.2".

6.7: Delete "HCFC Propellant" and substitute "Carbon dioxide propellant".

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* Add the following paragraph:

"6.9 Pollution prevention. Review of this specification in accordance with Executive Order 12856 has resulted in the replacement of xylene (listed in SD-14) with aliphatic naphtha in 4.7.5.1."

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* Figure 1: Delete title in entirety and substitute "Acid salt fog protection test apparatus."

NOTE: The margins of this amendment are marked with asterisks to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

CONCLUDING MATERIAL

Custodians:
Army - EA
Navy - AS
Air Force - 99

Review Activities:
Army - AT, MR
Navy - CG, OS
Air Force - 84

Preparing activity:
Navy - AS
(Project 8030-0743)

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